Dimitris Volteras, PhD

Computational Biologist | Machine Learning Scientist

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PROFILE

Computational biologist and machine learning scientist developing predictive models of complex cellular systems from high-dimensional biological data. My research lies at the intersection of modern machine learning and computational biology, spanning generative and representation learning, single-cell biology, dynamical modelling, and probabilistic approaches. I am particularly interested in cellular dynamics, gene regulation and perturbation response, and in how biological structure and experimental design can provide useful inductive biases for ML models. I work closely with experimental scientists to translate biological questions into computational models and testable hypotheses.

CORE EXPERTISE

Machine Learning & Mathematical Modelling: generative modelling | representation and self-supervised learning | dynamical models and neural ODEs/SDEs | Bayesian inference | causal and mechanistic modelling | stochastic processes | probabilistic modelling

Computational & Single-Cell Biology: single-cell transcriptomics (scRNA-seq) | perturbational and interventional data (Perturb-seq, signalling screens) | gene regulatory networks | cellular dynamics and trajectory inference | developmental biology | lineage tracing | metabolic RNA labelling

Programming & Computing: Python (PyTorch, scikit-learn, NumPy, SciPy, scanpy, scVI) | Julia | R | Mathematica | distributed and multi-GPU training | large-scale simulation | SLURM/HPC | Linux/Bash | Git/GitHub

EXPERIENCE

Postdoctoral Research Fellow | Francis Crick Institute, London

Developmental Dynamics Laboratory - Dr James Briscoe | Mar 2025 - Present

• Developed a **generative latent-variable model with continuous-time dynamics** to learn how gene regulatory networks transform signalling inputs into cell fate outcomes from large-scale, multi-condition single-cell signalling perturbation data.

• Developed a **probabilistic framework for inferring hierarchical lineage relationships** from single-cell genomic barcoding data, contributing to an interdisciplinary study of neural tube developmental dynamics (Nature, accepted).

• Led development of a **stochastic modelling and inference framework for gene regulatory mechanisms** from time-resolved single-cell transcriptomics, integrating metabolic RNA labelling and developmental time information.

• Built and optimised **scalable ML and numerical simulation pipelines** on HPC infrastructure, including distributed multi-GPU model training.

• Exploring **predictive representation-learning approaches for cellular perturbation response** from high-dimensional interventional single-cell data.

• Led a **team project in a Data Science Hackathon**, benchmarking dimensionality reduction methods for single-cell data.

• Supervised a visiting PhD student on stochastic modelling and inference of gene regulation from single-cell transcriptomics data.

Postdoctoral Research Associate | Imperial College London

Computational Molecular Systems Biology Group - Dr Vahid Shahrezaei | Sep 2024 - Feb 2025

• Developed a **stochastic agent-based model** to investigate the spatiotemporal dynamics of tumour progression.

• Contributed computational modelling and quantitative analysis to an interdisciplinary study identifying axonal injury as a targetable driver of glioblastoma progression (Nature, 2025).

PhD Researcher in Computational Biology | Imperial College London
Computational Molecular Systems Biology & Single-Cell Dynamics Groups - Dr Vahid Shahrezaei & Dr Philipp Thomas | Oct 2020 - Sep 2024

• Developed **stochastic and probabilistic methods to infer gene regulatory mechanisms** from high-dimensional single-cell transcriptomics.
• Developed a **Bayesian inference framework for genome-wide transcriptional dynamics** from time-resolved single-cell RNA sequencing, published in Cell Systems.
• Investigated how **cell-to-cell variability and dynamical correlations** can reveal modes of gene regulation from single-cell measurements.
• Co-supervised **MSc and BSc research projects on gene regulatory network inference** from single-cell transcriptomics.
• Taught and assessed undergraduate and MSc courses in probability, data science, applied mathematics and systems biology.

EDUCATION

PhD in Applied Mathematics & Computational Biology | Imperial College London | 2020 - 2024
Thesis: “Data-driven stochastic modelling of gene regulation” - Roth PhD Scholarship, Department of Mathematics

MSc in Applied Mathematics | Imperial College London | 2019-2020
Thesis: “Multiscale approximations of stochastic reaction networks” - Focus: Mathematical biology, Machine learning, Stochastic differential equations, Markov processes

BSc in Mathematics | National and Kapodistrian University of Athens | 2014-2018

Focus: Probability theory, Stochastic analysis, Bayesian inference, ODEs and PDEs, and Linear algebra.

SELECTED PUBLICATIONS AND MANUSCRIPTS

- Volteras, D., Thomas, P., & Shahrezaei, V. (2024). Global transcription regulation revealed from dynamical correlations in time-resolved single-cell RNA sequencing. Cell Systems. DOI: 10.1016/j.cels.2024.07.002
- Boezio, G. L. M., Depotter, J. R. L., Frith, T. J. R., Radley, A., Volteras, D., Strohbuecker, S., et al. (2026). Hierarchical lineage architecture of human and avian spinal cord revealed by single-cell genomic barcoding. Nature. Accepted.
- Clements, M., Tang, W., Baronik, Z. F., Ragdale, S., Oria, R., Volteras, D., White, I. J., et al. (2025). Axonal injury is a targetable driver of glioblastoma progression. Nature. DOI: 10.1038/s41586-025-09411-2
- Volteras, D., Thomas, P., & Shahrezaei, V. Inferring causal gene regulatory relationships from time-resolved single-cell transcriptomics with mechanistic machine learning. Manuscript in preparation.
- Lapostolle, B., Amad, H., Volteras, D., et al. Flow maps for trajectory inference. Manuscript in preparation.

SELECTED CONFERENCE TALKS

- Invited talk - European Conference on Mathematical & Theoretical Biology, Graz (2026)
- Invited talk - Data-Driven Modelling Workshop, Imperial College London (2026)
- Contributed talk - Causality in Biology & AI Symposium, Barcelona (2025)
- Contributed talk - AI x BIO, Wellcome Genome Campus, Cambridge (2025)
- Contributed talk - Single Cell Biology, Wellcome Genome Campus, Cambridge (2024)
- Contributed talk - Single Cell Analyses, Cold Spring Harbor Laboratory, New York (2023)

SELECTED AWARDS

- Best Project Award, AI Hackathon co-organised by Google DeepMind and the Francis Crick Institute (2026)
- Roth PhD Scholarship, Department of Mathematics, Imperial College London (2020)